Taha Irfan

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Education

University of Guelph Sep 2024 – Present
B.Eng, Computer Engineering (Co-op)

- **Relevant Coursework:** Large-Scale Software Architecture, Data Structures & Algorithms, Computer Organization and Architecture.

Skills

Languages: Python, Java, C, SQL (SQLite), VHDL
Python Tools: pandas, sqlite3, pyserial, openpyxl, pyreverse
Testing & Debugging: TestFlight, Jira, Root-Cause Analysis, Automated Test Harnesses, ARM DWT Cycle Profiling
Modelling: UML (Class, Use Case, Activity, Package Diagrams), PlantUML, Mermaid
Tools & Platforms: Git/GitHub, VS Code, Linux, JavaFX, STM32CubeIDE, KiCad, Claude Code

Experience

QA / Testing Intern | Moves (Tech Startup) May 2026 – Aug 2026

- Tested pre-release builds of the team's iOS social app through **TestFlight**, covering each new feature before launch.
- Logged every defect as a **Jira** ticket with reproducible steps and routed it to the developer to fix.
- Reproduced bugs through **root-cause analysis**, isolating each trigger before the app launched that summer.

Projects

HIL-Testbench – Automated Test Framework | ***Python, C, pyserial, STM32***

- To measure firmware response precisely, built a **Python/pyserial** harness driving **C** firmware, running 10,000 tests.
- Measured a steady **642 ns** response with the DWT cycle counter, separating serial overhead (2.67 ms vs. 2 ms expected).
- Traced a rare failure in **0.06%** of tests to a background interrupt, fixed it, and documented the root cause.

Temperature Logging Data Pipeline | ***Python, SQLite, pandas, KiCad***

- To analyze sensor data before the hardware was ready, built a **Python** pipeline storing simulated readings in **SQLite**.
- Wrote **SQL** queries for minimum, maximum and average readings, exporting data and a summary to Excel with **pandas**.
- Designed the 2-layer **STM32** sensor board in **KiCad** that will stream real readings into the pipeline over UART.

AI Outreach Automation Tool | ***Claude Code, LLMs***

- To reach engineers at target companies for coffee chats, built a **Claude Code** tool that automates first-contact emails.
- Ranked target companies into three priority tiers, so the tool reaches the best-fit contacts first.
- Limited automation to first messages, keeping every follow-up personal.

Leadership & Extracurriculars

Chair | Guelph Engineering Society Jan 2025 – Apr 2025

- Facilitated weekly meetings and set agendas as a first-year student among upper-year students and faculty.
- Settled a budget dispute between two clubs by having each side back its case with **data**, then holding a vote.
- Resolved students' concerns by working with stakeholders through grievance committees.

Open-Source Contributor | beets Music Manager (Team of 7) Sep 2026 – Present

- Ran the team's **Jira** Scrum board linked to **GitHub**, tracing every task from plan to merged code.
- Turned two user-reported bugs into **use case specifications**; the team merged both fixes after testing.

Certifications

Cisco IT Essentials – Cisco Networking Academy
Certified SOLIDWORKS Associate (CSWA) – Dassault Systèmes

Interests

Building custom PCs, short-form video editing (YouTube Short with 10,000+ views), embedded hardware design.